RACE Regular Analysis Through Continuous Exercise

Solution of Triangles: RACE-01

Single Correct Question

[**Note:** All symbols used have usual meaning in triangle.]

1. The real value of x such that $x^2+2x, 2x+3$ and x^2+3x+8 are lengths of the sides of a triangle is equal to

(A) $(5,\infty)$ (B) $(4,\infty)$ (C) $(3,\infty)$ (D) $(0,\infty)$

MTS117

2. In a ΔABC, if $\angle A=\frac{\pi}{4}, \angle B=\frac{5\pi}{12}$, then $\frac{a+c\sqrt{2}}{b}$ is equal to

(A) 1 (B) 2 (C) 3 (D) 4

MTS118

3. If in a ΔABC, $\angle A=\frac{\pi}{6}$ and $b:c=2:\sqrt{3}$, then $\angle B=$

(A) 30° (B) 45° (C) 90° (D) 60°

MTS119

4. In any ΔABC. $a\sin(B-C)+b\sin(C-A)+c\sin(A-B)=$

(A) 1 (B) 0 (C) 3 (D) 5

MTS120

Rankers Academy JEE

5. The perimeter of a triangle ABC is six times the arithmetic mean of the sines of its angles. If the side b is equal to $\sqrt{3}$, then angle $B=$

(A) 30° (B) 45° (C) 90° (D) 60°

MTS121

6. If $a:b:c=4:5:6$, and $\angle C=\lambda\angle A$, then $\lambda=$

(A) 1 (B) 2 (C) 3 (D) 4

MTS122

7. In a ΔABC, if $9a^2+9b^2=19c^2$, then the value of $\frac{\cot C}{\cot A+\cot B}$ is equal to

(A) $\frac{1}{9}$ (B) $\frac{2}{9}$ (C) $\frac{5}{9}$ (D) $\frac{7}{9}$

MTS123

8. In a triangle ABC, the sides a,b,c are roots of $x^3-11x^2+38x-40=0$. If $\sum\left(\frac{\cos A}{a}\right)=\frac{p}{q}$, then the least value of $(p+q)$ where $p,q\in N$ is equal to

(A) 10 (B) 15 (C) 20 (D) 25

MTS124

For More Join: @IITJEE_Advanced

9. In a triangle $ABC, a : b : c = 4 : 5 : 6$. Then $3A + B$ equals to

(A) $4C$ (B) 2π (C) $\pi - C$ (D) π

MTS125

10. If in a triangle ABC. $\sin A, \sin B$ & $\sin C$ are in A.P. $(C > 90°)$, then $\cos A$ is

(A) $\frac{3c-4b}{2b}$ (B) $\frac{3c-4b}{2c}$ (C) $\frac{4c-3b}{2b}$ (D) $\frac{4c-3b}{2c}$

MTS126

Subjective Type Question

11. In any triangle ABC, if $\frac{a^2-b^2}{a^2+b^2}=\frac{\sin(A-B)}{\sin(A+B)}$, then prove that the triangle is either right angled or isosceles.

MTS127

12. If the angles of ΔABC are in the ratio 1 : 2 : 3, then find the ratio of their corresponding sides

MTS128

13. In any ΔABC, prove that $\frac{b-c}{b+c}=\frac{\tan\left(\frac{B-C}{2}\right)}{\tan\left(\frac{B+C}{2}\right)}$.

MTS129

14. If in a triangle $ABC, \angle C = 60°$, then prove that $\frac{1}{a+c}+\frac{1}{b+c}=\frac{3}{a+b+c}$.

MTS130

15. In a ΔABC prove that $\frac{c}{a-b}=\frac{\tan\frac{A}{2}+\tan\frac{B}{2}}{\tan\frac{A}{2}-\tan\frac{B}{2}}$

MTS131

16. In ΔABC, if A, B, C are in A.P., then prove that $2\cos\frac{A-C}{2}=\frac{a+c}{\sqrt{a^2+c^2-ac}}$.

MTS132

Rankers' Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	B	C	B	D	B	C	D	D	D

12. $a : b : c = \sin A : \sin B : \sin C = 1 : \sqrt{3} : 2$

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Solution of Triangles: RACE-02

Single Correct Question

[**Note:** All symbols used have usual meaning in triangle.]

1. In ΔABC, the value of $\frac{\cos A+\cos C}{a+c}+\frac{\cos B}{b}=$

(A) $\frac{1}{b}$ (B) $\frac{2}{b}$ (C) $\frac{3}{b}$ (D) $\frac{4}{b}$

MTS133

2. In ΔABC, the value of $a(b^2+c^2)\cos A+b(c^2+a^2)\cos B+c(a^2+b^2)\cos C=$

(A) abc (B) $2abc$

(C) $3abc$ (D) $4abc$

MTS134

3. In ΔABC, the value of $(b+c)\cos A+(c+a)\cos B+(a+b)\cos C=$

(A) $a-b+c$ (B) $a+b-c$

(C) $a-b-c$ (D) $a+b+c$

MTS135

Pankers Academy JEE

4. If in a ΔABC, $b=3c$, then the value of $\cot\frac{A}{2}.\cot\frac{B-C}{2}$ is equal to

(A) 1 (B) 2 (C) 4 (D) 6

MTS136

5. In a triangle ABC, $\angle A=60°$ and $b:c=\sqrt{3}+1:2$, then the value of $(\angle B-\angle C)$ is equal to

(A) 15° (B) 30° (C) 45° (D) 60°

MTS137

6. In any ΔABC, $(abcs)$ $\sin\frac{A}{2}.\sin\frac{B}{2}.\sin\frac{C}{2}=$

(A) Δ^2 (B) Δ (C) Δ^3 (D) $\frac{1}{\Delta}$

MTS138

7. If in a triangle ABC, $\Delta=a^2-(b-c)^2$, then the value of $\tan A$ =

(A) $\frac{2}{15}$ (B) $\frac{4}{15}$ (C) $\frac{8}{15}$ (D) $\frac{7}{15}$

MTS139

8. In a triangle ABC, if $\tan\frac{A}{2}.\tan\frac{B}{2}=\frac{1}{3}$ and $ab=4$, then the least value of c is equal to

(A) 1 (B) 2 (C) 3 (D) 5

MTS140

For More Join: @IITJEE_Advanced

9. In ΔABC, if a, b, c (taken in that order) are in A.P. then the value of $\cot\frac{A}{2}\cot\frac{C}{2}$ is equal to

(A) 5 (B) 7 (C) 3 (D) 9

MTS141

Subjective Type Question

10. In a ΔABC, prove that $\frac{\cos A}{c\cos B+b\cos C}+\frac{\cos B}{a\cos C+c\cos A}+\frac{\cos C}{a\cos B+b\cos A}=\frac{a^2+b^2+c^2}{2abc}$.

MTS142

11. ΔABC with $\angle A$ acute, $b = 2$ and $c=\left(\sqrt{3}-1\right)$ has area $\left(\sqrt{3}-1\right)/2$, then find the measures of angles of triangle.

MTS143

12. If A, B, C are the angle of a triangle, then prove that $\cot\frac{A}{2}+\cot\frac{B}{2}+\cot\frac{C}{2}=\frac{s^2}{\Delta}$

MTS144

13. In any triangle ABC, prove that $(a+b+c)\left(\tan\frac{A}{2}+\tan\frac{B}{2}\right)=2c\cot\frac{C}{2}$.

MTS145

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	A	C	D	B	B	A	C	B	C

11. $\angle A = 30°, \angle B = 135°, \angle C = 15°$

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Solution of Triangles: RACE-03

Single Correct Question:

[**Note:** All symbols used have usual meaning in triangle.]

1. In a ΔABC, the value of $r\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$ is equal to

(A) $\frac{\Delta}{R}$ (B) $\frac{\Delta}{2R}$ (C) $\frac{\Delta}{3R}$ (D) $\frac{\Delta}{4R}$

MTS146

2. If the median AD of a ΔABC makes an angle $3\pi/4$ with the side BC, then evaluate $|\cot B - \cot C|$.

(A) 4 (B) 2 (C) 1 (D) $\sqrt{3}$

MTS147

3. If in a ΔABC, $AB = 4\ cm, AC = 8\ cm$ and $\cos\left(\frac{B-C}{2}\right) = \frac{1}{\sqrt{2}}$; then evaluate $|2\cot B - \cot C|$.

(A) 1 (B) 2 (C) 3 (D) 4

MTS148

4. In ΔABC, if circumradius 'R' and inradius 'r' are connected by relation $R^2 - 4Rr + 8r^2 - 12r + 9 = 0$, then semi perimeter of the ΔABC is equal to

(A) $\frac{9\sqrt{3}}{2}$ (B) $\frac{5\sqrt{3}}{2}$ (C) $\frac{7\sqrt{3}}{2}$ (D) $\frac{\sqrt{3}}{2}$

MTS149

5. In a ΔABC, prove that $a\cos B\cos C + b\cos C\cos A + c\cos A\cos B =$

(A) $\frac{\Delta}{R}$ (B) $\frac{\Delta}{2R}$ (C) $\frac{2\Delta}{R}$ (D) $\frac{R}{\Delta}$

MTS150

6. The ratio of the area of n-sided regular polygon, circumscribed about a circle, to the area of the regular polygon of equal number of sides inscribed in the circle is 4 : 3. The value of n is equal to

(A) 3 (B) 6 (C) 9 (D) 10

MTS151

7. A regular pentagon and a regular decagon have the same perimeter, then their areas are in the ratio.

(A) $\frac{1}{\sqrt{5}}$ (B) $\frac{\sqrt{5}}{2}$ (C) $\frac{3}{\sqrt{5}}$ (D) $\frac{2}{\sqrt{5}}$

MTS152

8. If the area of circle is A_1 and area of regular pentagon inscribed in the circle is A_2, then $\frac{A_1}{A_2} =$

(A) $\frac{9\pi}{5\sqrt{10+2\sqrt{5}}}$ (B) $\frac{7\pi}{5\sqrt{10+2\sqrt{5}}}$ (C) $\frac{8\pi}{5\sqrt{10+2\sqrt{5}}}$ (D) $\frac{4\pi}{5\sqrt{10+2\sqrt{5}}}$

MTS153

For More Join: @IITJEE_Advanced

9. A circle is inscribed in a regular octagon whose area is Δ. If area of a new regular octagon formed by points of contact of polygon and incircle is Δ', then $\frac{\Delta}{\Delta'}$ is

(A) $\cos^2\frac{\pi}{8}$ (B) $\sec\frac{\pi}{8}$ (C) $\sec^2\frac{\pi}{8}$ (D) $\cot^2\frac{\pi}{8}$

MTS154

10. Let $A_i, i = 1,2,3,\ldots.10$ are vertices of regular decagon P. If area of polygon P is $5\sin\left(\frac{\pi}{5}\right)$ square units then the value of $(A_1A_2)(A_1A_3)(A_1A_4)\ldots.(A_1A_{10})$ is

(A) 10 (B) 9 (C) $45\sin\left(\frac{\pi}{5}\right)$ (D) $45\cos\left(\frac{\pi}{5}\right)$

MTS155

Subjective Type Question

11. Let Δ & Δ' denote the areas of a Δ and that of its incircle. Prove that

$\Delta : \Delta' = \left(\cot\frac{A}{2}.\cot\frac{B}{2}.\cot\frac{C}{2}\right) : \pi$

MTS156

12. In a ΔABC, prove that $\sin 2A + \sin 2B + \sin 2C = 2\Delta/R^2$

MTS157

13. In a ΔABC, prove that $\frac{b^2-c^2}{2a} = R\sin(B-C)$

Rankers Academy JEE

MTS158

14. In a ΔABC, show that $a+b+c = \frac{abc}{2Rr}$

MTS159

15. If the perimeter of a circle and a regular polygon of n sides are equal, then prove that

$\frac{\text{area of the circle}}{\text{area of polygon}} = \frac{\tan\frac{\pi}{n}}{\frac{\pi}{n}}$.

MTS160

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	B	C	A	A	B	D	C	C	A

For More Join: @IITJEE_Advanced